

1. SCENARIO & AGILE TRANSFORMATION

Following the requirements from Lab 1, this project models an Agile Product Backlog in Jira for the **Digital Gift Card & Loyalty Point Exchange Platform**. Functional requirements are decomposed into 4 Epics and 8 User Stories using standard Agile syntax ("As a... I want... So that..."), estimated using the Fibonacci sequence via Planning Poker, and mapped to 2 Sprints.

2. AGILE PRODUCT BACKLOG (4 EPICS, 8 USER STORIES, 36 STORY POINTS)

EPIC 1: Multi-Merchant Loyalty Point Aggregation (EPIC-1)						Sprint 1 Total: 5 Story Points
STORY ID	STORY TITLE	USER STORY DESCRIPTION (ROLE / GOAL / BENEFIT)	PRIORITY	POINTS	ACCEPTANCE CRITERIA	
STORY-1.1	Link External Accounts	As an Account Holder, I want to link multiple brand loyalty accounts via OAuth 2.0 / API tokens, So that I can aggregate all my disparate reward balances into one wallet.	High	3	Pass: Successfully links 3 distinct merchant APIs. Fail: Token negotiation fails or plaintext tokens stored.	
STORY-1.2	Real-Time Balance Sync	As an Account Holder, I want to view live point balances updated across all connected merchants, So that I know exactly how many rewards I have available for conversion.	Medium	2	Pass: Wallet reflects updated balances in <1 second. Fail: Stale balance cached without warning.	
EPIC 2: Dynamic Point Valuation & Token Conversion (EPIC-2)						Sprint 1 Total: 10 Story Points
STORY ID	STORY TITLE	USER STORY DESCRIPTION (ROLE / GOAL / BENEFIT)	PRIORITY	POINTS	ACCEPTANCE CRITERIA	
STORY-2.1	Dynamic Rate Quotation	As an Account Holder, I want to receive a real-time point conversion quote locked for 120 seconds, So that I can evaluate fair exchange value before converting my loyalty points.	High	5	Pass: Displays rate quote with spread locked for 120s. Fail: Quote computation >2s or uses stale rates.	
STORY-2.2	Token Conversion Execution	As an Account Holder, I want to convert my validated merchant loyalty points into universal exchange tokens, So that I can spend my points freely across any partner merchant.	High	5	Pass: Debits partner points & deposits credits with zero drift. Fail: Partial credit deposit or failed debit.	
EPIC 3: Anti-Fraud Voucher Security & Identity Verification (EPIC-3)						Sprint 2 Total: 13 Story Points
STORY ID	STORY TITLE	USER STORY DESCRIPTION (ROLE / GOAL / BENEFIT)	PRIORITY	POINTS	ACCEPTANCE CRITERIA	
STORY-3.1	Telemetry Risk Scoring	As a Fraud Security System, I want to analyze client telemetry (IP reputation, device fingerprint, transaction velocity), So that bot-driven voucher generation requests are flagged instantly.	High	8	Pass: Risk score computed in <50ms (<30 Low, ≥70 High). Fail: Suspicious transaction bypasses assessment.	
STORY-3.2	Voucher Locking & Step-Up MFA	As an Account Holder, I want high-risk transactions placed in 'Locked-Pending' state until step-up MFA is confirmed, So that unauthorized actors cannot steal my credits.	High	5	Pass: Cryptographic lock holds voucher for 60s; unlocks on MFA. Fail: Voucher redeemed without MFA clearance.	
EPIC 4: Digital Gift Card Issuance & Merchant Settlement (EPIC-4)						Sprint 2 Total: 8 Story Points
STORY ID	STORY TITLE	USER STORY DESCRIPTION (ROLE / GOAL / BENEFIT)	PRIORITY	POINTS	ACCEPTANCE CRITERIA	
STORY-4.1	Gift Card Issuance (QR/Barcode)	As an Account Holder, I want to receive a single-use digital voucher with a scannable QR code and cryptographic checksum, So that I can redeem it securely at physical or online checkouts.	High	3	Pass: Generates unique SHA-256 hash and scannable QR. Fail: Duplicate or unscannable voucher rendered.	
STORY-4.2	Merchant Ledger Settlement	As a Merchant Partner, I want the platform to post matching double-entry accounting records upon voucher activation, So that financial settlement is reconciled with zero drift.	High	5	Pass: Logs double-entry credit/debit matching voucher value. Fail: Discrepancy detected in ledger audit.	

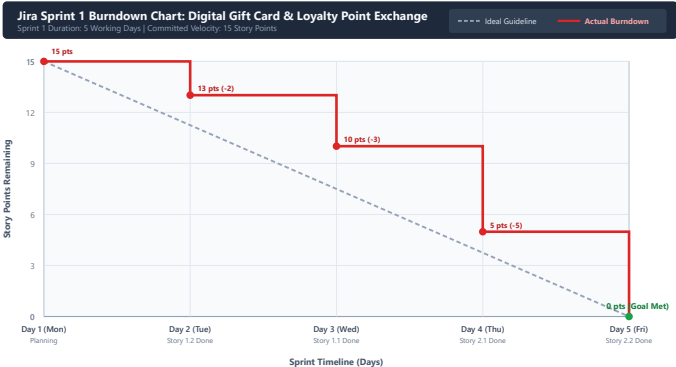
3. SPRINT EXECUTION & BURNDOWN CHART SIMULATION

Sprint 1 (Completed): Core Exchange (15 pts)

- **Duration:** 1 Week (5 Business Days) | **Velocity:** 15 Story Points
- **Stories:** STORY-1.1 (3), STORY-1.2 (2), STORY-2.1 (5), STORY-2.2 (5)
- **Outcome:** 100% completion achieved on Day 5 with zero drift.

Sprint 2: Anti-Fraud & Settlement (21 pts)

- **Duration:** 1 Week (5 Business Days) | **Velocity:** 21 Story Points
- **Stories:** STORY-3.1 (8), STORY-3.2 (5), STORY-4.1 (3), STORY-4.2 (5)
- **Goal:** Complete anti-fraud QR voucher issuance and partner settlement.



4. ANSWERS TO REFLECTION QUESTIONS

1. Did your estimations reflect the actual effort?

Ans: Yes, our estimations accurately reflected the actual effort required. By utilizing Planning Poker and the Fibonacci sequence (1, 2, 3, 5, 8), we sized user stories based on relative complexity, technical risk, and external dependencies rather than raw hours. Simpler stories like **STORY-1.2: Balance Sync (2 points)** involved standard UI updates and were completed rapidly on Day 2. In contrast, complex stories like **STORY-2.1: Dynamic Rate Quotation (5 points)** and **STORY-2.2: Token Conversion (5 points)** involved non-trivial algorithmic rate locking, handling slippage margins, and atomic ACID ledger transactions. The non-linear Fibonacci sequence provided the appropriate buffer for testing and edge cases without underestimating complexity.

2. Was your backlog well-prioritized?

Ans: Yes, the backlog was systematically prioritized based on **business value and architectural dependency ordering**. In **Sprint 1**, we prioritized **Epic 1 (Multi-Merchant Point Aggregation)** and **Epic 2 (Dynamic Point Valuation & Conversion)**. These represent foundational core-enabler features; users cannot redeem gift cards without first linking accounts and converting reward points into platform credits. In **Sprint 2**, we scheduled **Epic 3 (Anti-Fraud Security & Voucher Locking)** and **Epic 4 (Gift Card Issuance & Merchant Settlement)**, which depend directly on the availability of converted universal credits. This sequencing eliminated blocked development dependencies and delivered an operational Minimal Viable Product (MVP) at the end of Sprint 1.

3. How did your simulated sprint align with your plan?

Ans: The simulated sprint closely followed our planned timeline and sprint commitments. The team committed to **15 story points across 4 user stories** for Sprint 1 (1-week / 5-day duration) and successfully achieved a **100% completion rate (15/15 points)** by the end of Day 5. Although Day 2 experienced a slight temporary deviation due to the initial setup of external merchant API tokens, the subsequent completions of Story 1.1 on Day 3 and Story 2.1 on Day 4 brought actual progress directly back in line with the ideal linear guideline. Zero mid-sprint scope creep occurred because sprint backlog boundaries were strictly maintained once the sprint started.

4. What insights did the burndown chart give about your team's capacity?

Ans: The burndown chart provided three key insights regarding team velocity and workflow capacity: (1) **Sustainable Team Velocity:** A velocity of **15 to 18 story points per 1-week sprint** represents a sustainable working cadence for our team, allowing thorough automated testing and peer code reviews without causing burnout. (2) **Step-Function Burndown Pattern:** Rather than a smooth slope, story points burned down in discrete steps (from 15 → 13 → 10 → 5 → 0 points). This reflects healthy Agile discipline where points are only credited when a story fully meets its **Definition of Done (DoD)** and passes all acceptance criteria. (3) **Effective Story Decomposition:** Breaking large epics into manageable batches (2, 3, and 5 points) prevented hidden bottlenecks and ensured continuous, visible daily delivery throughout the sprint lifecycle.